

gemstone-py Framework Adapters

Adding sync and async web framework adapters

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Adding a Framework Adapter

Use `gemstone_py.web_core` for request lifetime and transaction decisions. Adapters should stay thin: translate the framework's request hooks into `RequestScope` or `AsyncRequestScope`, then expose one helper that returns the current request session.

Sync Framework Shape

```
from gemstone_py import GemStoneConfig, RequestScope, TransactionPolicy
from gemstone_py.session_providers import GemStoneSessionPool

pool = GemStoneSessionPool(maxsize=4, config=GemStoneConfig.from_env())

def begin_request(request):
    request.gemstone_scope = RequestScope(
        session_provider=pool,
        transaction_policy=TransactionPolicy.COMMIT_ON_SUCCESS,
    )

def current_session(request):
    return request.gemstone_scope.session()

def end_request(request, exc=None, response_status=None):
    request.gemstone_scope.finalize(exc, response_status=response_status)
```

The scope commits on successful responses, aborts on exceptions or server-error status codes, and releases pooled sessions with the correct `clean` / `discard` flags.

Async Framework Shape

```
from gemstone_py import AsyncRequestScope, GemStoneConfig, TransactionPolicy
from gemstone_py.aio.pool import AsyncSessionPool

pool = AsyncSessionPool(maxsize=8, config=GemStoneConfig.from_env())

async def begin_request(scope):
    scope["gemstone_scope"] = AsyncRequestScope(
        session_provider=pool,
        transaction_policy=TransactionPolicy.COMMIT_ON_SUCCESS,
    )
    scope["gemstone_session"] = await scope["gemstone_scope"].session()
```

```
async def end_request(scope, exc=None, response_status=None):
    await scope["gemstone_scope"].finalize(exc, response_status=response_status)
```

Existing Adapters

Framework	Module	Pattern
Flask	<code>gemstone_py.frameworks.flask</code>	request-local sync session
Django	<code>gemstone_py.frameworks.django</code>	middleware-managed sync session
FastAPI	<code>gemstone_py.aio.fastapi</code>	async dependency and ASGI middleware
Litestar	<code>gemstone_py.aio.litestar</code>	async dependency and ASGI middleware

Keep optional framework imports inside app/example code or function bodies, not at package import time. The adapter module should remain importable in a plain `gemstone-py` installation.

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